

ALINA SHAJI

B.Tech CSE – AI & ML Specialisation . Honours in Cyber Security | Batch 2023–27 | CHRIST (Deemed to be University), Bengaluru
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PROFESSIONAL SUMMARY

Final year B.Tech Computer Science Engineering student at CHRIST (Deemed to be University), specializing in Artificial Intelligence and Machine Learning with an Honours track in Cybersecurity. Experienced in designing and deploying end-to-end AI/ML systems including computer vision pipelines, intelligent browser security tools, and LLM-based applications. Proficient in Python, machine learning frameworks, and RESTful API development, with hands-on exposure to adversarial machine learning, threat detection, and retrieval-augmented generation. Demonstrated ability to translate research prototypes into production-ready solutions with strong analytical, problem-solving, and cross-functional collaboration skills. Seeking a software engineering or AI/ML role at a technology-driven organization where innovation and security-conscious engineering deliver measurable impact.

WORK EXPERIENCE & INTERNSHIPS

MatryxSoft Tech LLP, Bengaluru

Apr 2026 – May 2026

AI & Cyber Security Intern, QA Testing Intern

- Engineered SentinelX, an intelligent Chrome extension delivering real-time protection against phishing, insecure connections, and malicious web activity through continuous traffic interception and behavioral analysis.
- Developed a multi-signal threat detection engine integrating URL entropy analysis, domain reputation scoring, and heuristic classifiers to flag high-risk sites prior to page load reducing user exposure to zero-day phishing campaigns.
- Reduced false-positive rates by maintaining a curated high-reputation domain allowlist and implementing session-level caching, optimizing performance across repeat visits without compromising detection coverage.
- Designed an accessible UI with real-time risk scores and contextual security alerts, balancing usability for general users with the precision and transparency expected by security-conscious professionals.
- Executed end-to-end QA and security testing for the company's web and mobile applications developing structured test plans, defining test cases aligned to functional and security requirements, and tracking defect resolution through to closure.
- Identified, documented, and triaged security vulnerabilities and functional defects across test cycles; coordinated findings with the development team to ensure timely remediation and on-schedule delivery of all project milestones.
- Demonstrated strong risk assessment and attention to detail by systematically prioritizing high-severity issues, maintaining comprehensive testing documentation, and ensuring zero critical defects escaped to production.

MatryxSoft Tech LLP, Bengaluru

Apr 2025 – May 2025

Computer Vision & AI Intern

- Designed and implemented a real-time hand gesture recognition system mapping predefined gestures to system-level application launch commands, eliminating dependency on traditional keyboard and mouse input peripherals.
- Curated and annotated a custom gesture dataset from scratch; trained and fine-tuned a classification model achieving consistent recognition accuracy across varied lighting conditions, hand sizes, and skin tones.
- Integrated OpenCV-based video processing with a Flask backend, enabling sub-second gesture-to-action response times validated across live deployment environments.

PROJECTS

SentinelX – Intelligent Threat Detection Browser Extension | JavaScript, Chrome APIs, ML Heuristics, CSS

- Built a production-deployed browser extension that proactively intercepts and analyzes web requests, flagging phishing sites, malicious URLs, and insecure connections before the user interacts with any content.
- Engineered a multi-layered scoring engine using entropy analysis, domain age heuristics, and behavioral signals with a persistent trusted-domain registry achieving low false-positive rates without compromising threat sensitivity.
- Implemented a one-shot audio alert system with mute persistence via chrome.storage and a dark-mode threat card UI providing clear, real-time risk communication to end users.

Hand Gesture–Based Application Launcher | Python, OpenCV, MediaPipe, Flask, Custom Dataset

- Developed an end-to-end gesture-controlled launcher spanning raw video capture, MediaPipe landmark extraction, classification, and OS-level application triggering with zero physical input required.
- Constructed and labeled a custom gesture dataset; trained a classifier generalizing across hand sizes, skin tones, and environmental lighting validated across multiple test environments.
- Deployed a Flask execution backend translating gesture events into system commands with minimal latency, demonstrated in live environments.

LLM-Powered RAG System | Python, LangChain, FAISS, OpenAI API, Flask

- Designed a Retrieval-Augmented Generation pipeline grounding large language model responses in user-supplied documents, substantially reducing hallucination on domain-specific factual queries.

- Implemented document chunking, vector embedding via FAISS, and semantic similarity retrieval to surface contextually relevant passages before generation.
- Benchmarked the RAG pipeline against a baseline LLM, demonstrating measurably higher factual precision on structured document Q&A tasks.

SKILLS

Technical:	Python, Java, C, JavaScript, HTML, CSS, MySQL, FAISS (Vector Store), Git, GitHub, Figma, VS Code
AI / ML:	Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Model Training & Evaluation, Retrieval-Augmented Generation, Adversarial ML
Cybersecurity:	Threat Detection, Vulnerability Assessment, Browser Security, Phishing Analysis, Security Heuristics, Red-Teaming Fundamentals
Frameworks & Tools:	Flask, FastAPI, Streamlit, OpenCV, LangChain, LangGraph, MediaPipe, ChromaDB, Tailwind CSS, REST APIs
Soft Skills:	Analytical Problem Solving, Research-to-Deployment Execution, Technical Communication, Cross-functional Collaboration
Languages:	English — Proficient (C1) Kannada, Malayalam — Fluent Hindi — Conversational

CERTIFICATIONS & TRAINING

- **Generative AI** — NVIDIA Deep Learning Institute
- **Data Science** — Microsoft
- **Build a Chatbot** — IBM
- **Internet of Things (IoT)** — IIT Madras — NPTEL
- **HTML5 Application Development** — Infosys Springboard
- **Java Fundamentals** — Oracle Academy
- **Back-End Development** — Meta

PUBLICATIONS

Book Chapter | Auerbach Publications (Taylor & Francis Group), 2026

“Heart Disease Prediction Analysis Using Quantum-Enhanced Features with Classical and Quantum Machine Learning Models”

In: *Quantum Machine Learning*, 1st Edition — Auerbach Publications / Taylor & Francis Group.

- Co-authored a peer-reviewed book chapter investigating quantum-enhanced feature engineering for cardiovascular disease prediction benchmarking classical ML models against quantum ML counterparts to evaluate predictive accuracy and computational advantage.
- Applied quantum feature maps and variational quantum circuits to a clinical heart disease dataset, demonstrating statistically significant improvement in classification performance over baseline classical models.
- Published in a Taylor & Francis Group title establishing early research credibility at the intersection of quantum computing, healthcare AI, and predictive analytics.

CO-CURRICULAR & LEADERSHIP

Department of Computer Science, CHRIST University, Bengaluru

2023– Present

Event Coordinator, Organizer and Volunteer

- Coordinated technical workshops and coding events across the department, managing logistics, speaker coordination, and participant communication end-to-end.
- Collaborated with faculty and student committees to design event formats aligned with industry skill requirements, including sessions on AI/ML tools and cybersecurity fundamentals.

EDUCATION

CHRIST (Deemed to be University), Bengaluru

Jul 2023 – May 2027 (Expected)

B.Tech — Computer Science & Engineering | AI & ML Specialisation | Honours: Cyber Security | CGPA: 8.92 / 10.0

St. Claret Pre-University College, Bengaluru

Apr 2023

Class XII — CBSE | 84%

SSM Public School, Bengaluru

Apr 2021

Class X — ICSE | 90%